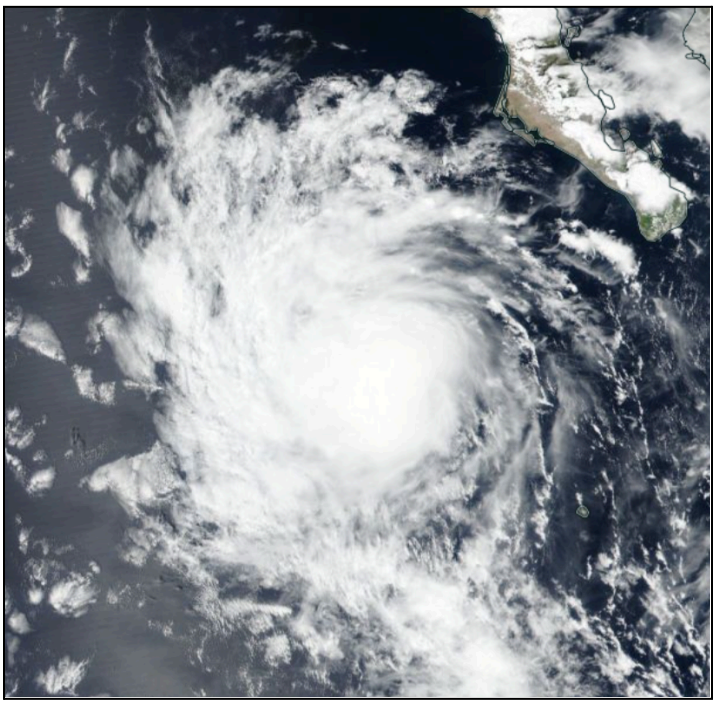
NATIONAL HURRICANE CENTER TROPICAL CYCLONE REPORT

TROPICAL STORM MARIO

(EP132025)

11–16 September 2025

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National Hurricane Center
16 December 2025



SUOMI NPP VIIRS SATELLITE IMAGE OF TROPICAL STORM MARIO AT 2056 UTC 15 SEPTEMBER 2025, AROUND THE TIME OF PEAK INTENSITY. IMAGE COURTESY OF NASA WORLDVIEW.

Mario produced gusty winds and heavy rainfall when it passed just offshore of the coast of southwestern Mexico as a small tropical storm. After briefly degenerating into a trough, Mario redeveloped into a stronger tropical storm that moved over open waters well offshore of the Baja California peninsula.

Tropical Storm Mario

11–16 September 2025

SYNOPTIC HISTORY

Mario appears to have originated from a tropical wave that crossed Central America and emerged over the eastern Pacific Ocean on 8–9 September. The wave produced disorganized showers and thunderstorms while moving westward, and a broad area of low pressure formed in association with the wave on 10 September, a couple of hundred n mi south of the coast of southern Mexico. The broad low turned northwestward while convective activity increased and gradually became better organized. Passive microwave data and satellite images indicate a well-defined, low-level circulation center formed early on 11 September, and it is estimated that a tropical depression formed by 0600 UTC that day, about 160 n mi southeast of Acapulco, Mexico. After briefly moving north-northwestward, the depression turned west-northwestward later on 11 September while moving around a low- to mid-level ridge over northern Mexico and Texas. The depression strengthened into Tropical Storm Mario by 0000 UTC 12 September, about 70 n mi southeast of Acapulco. The "best track" chart of Mario's path is given in Fig. 1, with the wind and pressure histories shown in Figs. 2 and 3, respectively. The best track positions and intensities are listed in Table 1[1].

Satellite images, surface observations, and microwave data show that the circulation of Mario was very small on 12 September while the tropical storm passed near the southwestern coast of Mexico (Fig. 4). Convection became less organized later that day due to the negative effects of moderate northeasterly shear, dry air entrainment, and the storm's close proximity to land. The small circulation became less defined and elongated in satellite images and scatterometer data, and Mario opened into a surface trough by 0000 UTC 13 September. Some deep convection flared up along the trough axis that morning while the remnants of Mario moved westward and interacted with a mesoscale convective system that moved off the coast of west-central Mexico. By midday, scatterometer data revealed the presence of a small but well-defined area of low pressure that formed in association with the remnants of Mario. This low moved west-southwestward to westward while being steered by a ridge to the north. The associated convection was disorganized late on 13 September, but increased in coverage and organization early the following morning. Mario became a tropical depression again by 0600 UTC 14 September, then strengthened into a tropical storm 6 h later when it was located about 270 n mi south of the southern tip of the Baja California peninsula.

Mario moved west-northwestward to northwestward during the next couple of days within the flow between a deep-layer ridge over northern Mexico and an upper-level trough offshore of the Baja California peninsula. Warm waters, weak vertical wind shear, and a moist mid-level

[1] A digital record of the complete best track, including wind radii, can be found on line at ftp://ftp.nhc.noaa.gov/atcf. Data for the current year's storms are located in the *btk* directory, while previous years' data are located in the *archive* directory.

environment supported strengthening on 14–15 September, and Mario reached a peak intensity of 55 kt at 1200 UTC 15 September, when it was located about 245 n mi southwest of the southern tip of the Baja California peninsula (cover image, Fig. 5). Soon after, Mario encountered cooler waters and moved into a drier and higher shear environment, which induced weakening as its deep convection quickly collapsed. Mario degenerated to a 35-kt post-tropical low by 1200 UTC 16 September and continued to weaken over the next couple of days while moving northwestward and gradually decelerating. The low degenerated to a trough of low pressure by 1200 UTC 18 September, roughly 400 n mi west of Cabo San Lazaro, Baja California Sur.

METEOROLOGICAL STATISTICS

Observations in Mario (Figs. 2 and 3) include subjective satellite-based Dvorak technique intensity estimates from the Tropical Analysis and Forecast Branch (TAFB) and the Satellite Analysis Branch (SAB), objective Advanced Dvorak Technique (ADT) estimates and Satellite Consensus (SATCON) estimates from the Cooperative Institute for Meteorological Satellite Studies/University of Wisconsin-Madison. Data and imagery from NOAA polar-orbiting satellites including the Advanced Microwave Sounding Unit (AMSU), the NASA Global Precipitation Mission (GPM), the European Space Agency's Advanced Scatterometer (ASCAT), the Defense Meteorological Satellite Program (DMSP) satellites, and the Time-Resolved Observations of Precipitation structure and storm Intensity with a Constellation of Smallsats (TROPICS) satellites, among others, were also useful in constructing the best track of Mario.

There were no reports of sustained tropical-storm-force winds associated with Mario.

Winds and Pressure

The estimated peak intensity of Mario is 55 kt from 1200–1800 UTC 15 September. This is consistent with UW-CIMSS SATCON and AiDT objective estimates during this period, as well as a 3.5/55 kt subjective Dvorak classification from TAFB at 1800 UTC that day. Due to the relatively large spread in satellite intensity estimates (45–65 kt) on 15 September and a lack of scatterometer data around the time of peak intensity, this estimate is more uncertain than usual. The estimated minimum central pressure of 994 mb is based on the Knaff-Zehr-Courtney (KZC) pressure-wind relationship.

As Mario passed near the coast of southern Mexico, the Acapulco International Airport (MMAA) reported a peak sustained wind of 22 kt and a gust of 32 kt at 0315 UTC 12 September. A couple of days later, an automated Mexican Navy weather station on Socorro Island recorded a peak wind gust of 29 kt at 1730 UTC 14 September and a minimum sea level pressure of 1001.6 mb just before 0000 UTC 15 September as Mario passed near the island. The following day, an inland weather station at Ciudad Constitución in Baja California Sur reported a sustained wind of 21 kt and a gust of 37 kt at 1610 UTC 15 September while Mario passed well offshore.

Rainfall and Flooding

Mario and its remnants contributed to an extended period of rainfall that occurred across portions of southwestern and west-central Mexico and the Baja California peninsula from 11–16 September (Fig. 6). The heaviest rainfall occurred on 12–13 September when Mario made its closest approach to the southwestern coast of Mexico, and a couple of stations in Guerrero and Colima reported more than 6 inches (~150 mm) of rainfall during the 24-h period. Media reports[2] indicate some localized flooding occurred in parts of Guerrero, including Acapulco and Zihuatanejo.

Maximum reported Mexico rainfall totals by state from 11–16 September:

Guerrero: 9.06 inches (230 mm) at Laguna de Coyuca, Coyuca de Benítez
Nayarit: 8.58 inches (218 mm) at Gaviotas, Bahía de Banderas
Colima: 7.06 inches (179.2 mm) at P. deriv. Gregorio T. Q. Jala, Tecomán
Jalisco: 5.79 inches (147 mm) at Cajón de Peña, Tomatlán
Michoacán: 4.56 inches (115.8 mm) at Observatorio de Zamora, Zamora

After Mario degenerated to a remnant low, a large plume of residual mid-level moisture was advected northward by an offshore trough during the ensuing few days. Moist ascent over the mountains of the Baja California peninsula resulted in heavy rainfall that produced flash and river flooding in parts of Baja California Sur on 16–17 September, most notably affecting the town of San Ignacio. This lingering moisture also contributed to flooding rains across portions of the southwestern United States on 17–18 September.

CASUALTY AND DAMAGE STATISTICS

There were no direct fatalities[3] associated with Mario as a tropical cyclone. Media reports[4] indicate one indirect fatality occurred on 16 September in San Ignacio, Baja California Sur, where a 37-year-old man died by electrocution during flooding. There are no official damage estimates available at the time of this report.

No major damage was reported in the state of Guerrero, where Mario made its closest approach to the coast of Mexico. News articles indicate that gusty winds downed some trees and power poles in Acapulco, Zihuatanejo, and surrounding areas, which resulted in brief power and communication outages. Some landslides were also reported in Acapulco. More significant damage occurred in parts of the Mulegé and La Paz municipalities of Baja California Sur as a

[2] https://www.eluniversal.com.mx/estados/saldo-blanco-en-guerrero-por-paso-de-tormenta-tropical/
[3] Deaths occurring as a direct result of the forces of the tropical cyclone are referred to as “direct” deaths. These would include those persons who drowned in storm surge, rough seas, rip currents, and freshwater floods. Direct deaths also include casualties resulting from lightning and wind-related events (e.g., collapsing structures). Deaths occurring from such factors as heart attacks, house fires, electrocutions from downed power lines, vehicle accidents on wet roads, etc., are considered “indirect” deaths.
[4] https://codigonaranja.mx/electrocutado-en-san-ignacio-durante-emergencia-por-inundaciones-cobro-la-vida-de-un-hombre-de-37-anos/

result of flash and river flooding that was indirectly related to distant post-tropical Mario. More than 60 homes were damaged and a portion of the Transpeninsular Highway was washed out in San Ignacio.[5]

FORECAST AND WARNING CRITIQUE

Genesis

The first genesis of Mario was not forecast well in advance (Table 2a). The incipient wave was introduced in the Tropical Weather Outlook (TWO) only 72 h prior to formation, and it reached the medium and high categories 60 h and 54 h before genesis, respectively. The disturbance was introduced with a low chance of development in the 2-day TWO 60 h before genesis. The 2-day probabilities were raised to the medium and high categories 36 h and 12 h before formation, respectively. In general, the global models did not provide a very strong signal for genesis until only a few days before Mario formed.

The second genesis of Mario was not anticipated (Table 2b). The remnants of Mario were introduced with a low chance of formation in the 7-day (2-day) TWO only 18 h (12 h) prior to genesis. The 2-day probabilities did not reach the medium or high category before genesis was determined to have occurred in post analysis. Forecaster confidence for genesis was low because there were multiple vorticity centers present (including the remnants of Mario) within a broader gyre well south of Baja California. From the available model guidance, it was unclear whether these features would consolidate or remain separate, weaker features embedded within the broader circulation. Ultimately, the redevelopment that occurred was from the low-level vorticity associated with the remnants of Mario.

The location of Mario's initial genesis was accurately forecast (Fig. 7), with all of the formation areas capturing the region where Mario formed. The genesis areas for Mario's redevelopment were also accurate (Fig. 8), albeit with limited lead time.

Track

A verification of NHC official track forecasts for Mario is given in Table 3a. Official track forecast errors were lower than the mean official errors for the previous 5-yr period at 12–24 h and 72 h, but greater than the 5-yr means at 36–60 h and 96 h. Note that the forecast sample size is relatively small and irregularly distributed. This is due to a couple of factors: (1) Mario had a fairly short existence as a tropical cyclone, and (2) Mario briefly dissipated and is shown as a disturbance or low for about a day in the final best track. Current NHC verification practice only includes forecasts if the system was classified in the final best track as a tropical cyclone at both the forecast's initial time and at the projection's valid time.

A homogeneous comparison of the official track errors with selected guidance models is given in Table 3b. Since some models had difficulty tracking Mario due to its small size and/or showed it dissipating inland over Mexico, there are several models that did not have a large

[5] https://zetatijuana.com/2025/09/tormenta-deja-un-muerto-y-mas-de-60-familias-damnificadas/

enough sample size to be included in this comparison due to the homogeneity requirement (including the ECMWF, UKMET, COAMPS-TC, HMON, and Google DeepMind). Overall, the NHC track forecast performed well against the available guidance, with lower errors at most forecast times.

Intensity

A verification of NHC official intensity forecasts for Mario is given in Table 4a. Official intensity forecast errors were lower than the mean official errors for the previous 5-yr period at all forecast times, albeit for a relatively small sample size. A homogeneous comparison of the official intensity errors with available guidance models is given in Table 4b. The 12–36-h NHC intensity forecasts were better than almost every available model, but several models and consensus aids had lower errors at 48–96 h. Early NHC forecasts had a high bias and showed Mario becoming a hurricane, which ultimately did not occur.

Wind Watches and Warnings

A Tropical Storm Watch was issued by the government of Mexico at 1500 UTC 12 September for the southwestern coast of Mexico from Lazaro Cardenas to Punta San Telmo. The watch was never upgraded to a warning, and it was discontinued at 0300 UTC 13 September.

ACKNOWLEDGEMENTS

Dr. Philippe Papin (NHC) created the GTWO verification graphics (Figs. 7 and 8).

Table 1. Best track for Tropical Storm Mario, 11–16 September 2025.

Date/Time (UTC)	Latitude (°N)	Longitude (°W)	Pressure (mb)	Wind Speed (kt)	Stage
11 / 0000	14.0	97.9	1009	25	low
11 / 0600	14.7	98.0	1009	25	tropical depression
11 / 1200	15.3	98.1	1008	30	"
11 / 1800	15.5	98.3	1007	30	"
12 / 0000	16.0	99.1	1006	35	tropical storm
12 / 0600	16.5	100.2	1006	35	"
12 / 1200	17.0	101.4	1006	35	"
12 / 1800	17.3	102.5	1007	30	tropical depression
13 / 0000	17.8	104.0	1009	25	disturbance
13 / 0600	18.3	105.4	1009	25	"
13 / 1200	18.5	106.6	1009	25	low
13 / 1800	18.4	107.6	1008	30	"
14 / 0000	18.1	108.4	1008	30	"
14 / 0600	18.1	109.1	1007	30	tropical depression
14 / 1200	18.4	109.7	1005	35	tropical storm
14 / 1800	18.7	110.3	1003	40	"
15 / 0000	19.1	111.0	1000	45	"
15 / 0600	19.4	111.8	996	50	"
15 / 1200	19.9	112.8	994	55	"
15 / 1800	20.6	113.9	994	55	"
16 / 0000	21.1	114.7	996	50	"
16 / 0600	21.6	115.5	1000	45	"

Date/Time (UTC)	Latitude (°N)	Longitude (°W)	Pressure (mb)	Wind Speed (kt)	Stage
16 / 1200	22.0	116.3	1003	35	low
16 / 1800	22.6	117.3	1004	30	"
17 / 0000	23.3	118.0	1006	25	"
17 / 0600	23.9	118.6	1008	25	"
17 / 1200	24.5	119.1	1009	20	"
17 / 1800	24.8	119.4	1010	20	"
18 / 0000	24.9	119.5	1010	15	"
18 / 0600	25.0	119.6	1010	15	"
18 / 1200					dissipated
15 / 1200	19.9	112.8	994	55	minimum pressure & maximum winds

Table 2a. Number of hours in advance of Mario's initial formation associated with the first NHC Tropical Weather Outlook forecast in the indicated likelihood category. Note that the timings for the "Low" category do not include forecasts of a 0% chance of genesis.

	Hours Before Genesis	
	48-Hour Outlook	**168-Hour Outlook**
Low (<40%)	60	72
Medium (40%-60%)	36	60
High (>60%)	12	54

Table 2b. Number of hours in advance of Mario's second formation associated with the first NHC Tropical Weather Outlook forecast in the indicated likelihood category. Note that the timings for the "Low" category do not include forecasts of a 0% chance of genesis.

	Hours Before Genesis	
	48-Hour Outlook	**168-Hour Outlook**
Low (<40%)	12	18
Medium (40%-60%)	–	12
High (>60%)	–	–

Table 3a. NHC official (OFCL) and climatology-persistence skill baseline (OCD5) track forecast errors (n mi) for Tropical Storm Mario, 11–16 September 2025. Mean errors for the previous 5-yr period are shown for comparison. Official errors that are smaller than the 5-yr means are shown in boldface type.

	Forecast Period (h)							
	12	24	36	48	60	72	96	120
OFCL	**20.5**	**22.4**	58.3	77.5	67.5	**76.5**	134.7	
OCD5	46.1	98.8	121.8	134.2	201.1	217.1	340.3	
Forecasts	9	5	3	3	5	5	3	
OFCL (2020-24)	22.4	33.6	43.5	53.2	65.1	79.4	108.6	135.2
OCD5 (2020-24)	38.2	75.3	117.2	161.0	206.0	252.0	340.7	422.7

Table 3b. Homogeneous comparison of selected track forecast guidance models (in n mi) for Tropical Storm Mario, 11–16 September 2025. Errors smaller than the NHC official forecast are shown in boldface type. The number of official forecasts shown here will generally be smaller than that shown in Table 3a due to the homogeneity requirement.

Model ID	Forecast Period (h)							
	12	24	36	48	60	72	96	120
OFCL	21.0	19.5	66.9	77.5	67.5	76.5	134.7	
OCD5	46.5	106.9	119.6	134.2	201.1	217.1	340.3	
GFSI	21.6	19.9	91.1	120.1	119.2	128.5	183.3	
HWFI	28.9	49.5	**60.6**	**43.9**	75.0	91.0	175.1	
HFAI	33.5	29.2	97.7	112.7	113.2	102.2	162.8	
HFBI	37.2	23.1	91.8	95.8	72.6	**67.2**	**119.4**	
NVGI	55.8	91.8	102.4	100.0	118.3	131.3	173.8	
TVCE	23.8	**19.2**	77.7	88.0	78.2	89.9	173.1	
TVCX	23.5	20.2	77.7	88.0	78.2	89.9	173.1	
TVDG	23.7	19.6	75.2	90.5	82.9	95.5	175.3	
HCCA	22.6	23.2	75.5	**77.4**	83.5	102.8	192.4	
TABS	34.5	56.6	145.0	237.1	283.1	309.4	408.7	
TABM	32.1	56.6	106.5	152.8	195.3	192.2	254.5	
TABD	32.6	56.0	80.0	92.6	144.8	139.0	176.2	
Forecasts	8	4	2	3	5	5	3	

Table 4a. NHC official (OFCL) and climatology-persistence skill baseline (OCD5) intensity forecast errors (kt) for Tropical Storm Mario, 11–16 September 2025. Mean errors for the previous 5-yr period are shown for comparison. Official errors that are smaller than the 5-yr means are shown in boldface type.

	Forecast Period (h)							
	12	24	36	48	60	72	96	120
OFCL	**3.9**	**6.0**	**3.3**	**10.0**	**14.0**	**13.0**	**16.7**	
OCD5	6.0	9.4	10.3	15.7	11.0	5.2	12.3	
Forecasts	9	5	3	3	5	5	3	
OFCL (2020-24)	5.7	8.9	10.8	12.9	14.4	15.5	17.0	18.6
OCD5 (2020-24)	7.4	12.4	16.2	19.2	21.0	21.3	21.5	21.6

Table 4b. Homogeneous comparison of selected intensity forecast guidance models (in kt) for Tropical Storm Mario, 11–16 September 2025. Errors smaller than the NHC official forecast are shown in boldface type.

Model ID	Forecast Period (h)							
	12	24	36	48	60	72	96	120
OFCL	3.9	6.0	3.3	10.0	14.0	13.0	16.7	
OCD5	6.0	9.4	10.3	15.7	**11.0**	**5.2**	**12.3**	
HWFI	8.8	10.4	6.0	**8.0**	**13.8**	14.0	**9.7**	
HFAI	10.2	18.0	13.7	**7.3**	**11.0**	19.0	29.0	
HFBI	7.1	12.2	4.3	11.3	**8.6**	14.0	16.7	
DSHP	6.1	6.6	7.0	13.0	15.0	**9.6**	**7.0**	
LGEM	6.0	7.2	8.3	**6.0**	**7.2**	**7.6**	**8.0**	
IVCN	5.9	9.0	6.7	**6.0**	**5.4**	**7.0**	**10.0**	
IVDR	6.6	9.2	6.7	**5.7**	**6.2**	**9.2**	**13.7**	
GFSI	8.0	10.6	10.3	12.7	18.4	29.4	33.7	
HCCA	5.1	**5.6**	4.3	**9.7**	**8.4**	**10.4**	22.3	
Forecasts	9	5	3	3	5	5	3	

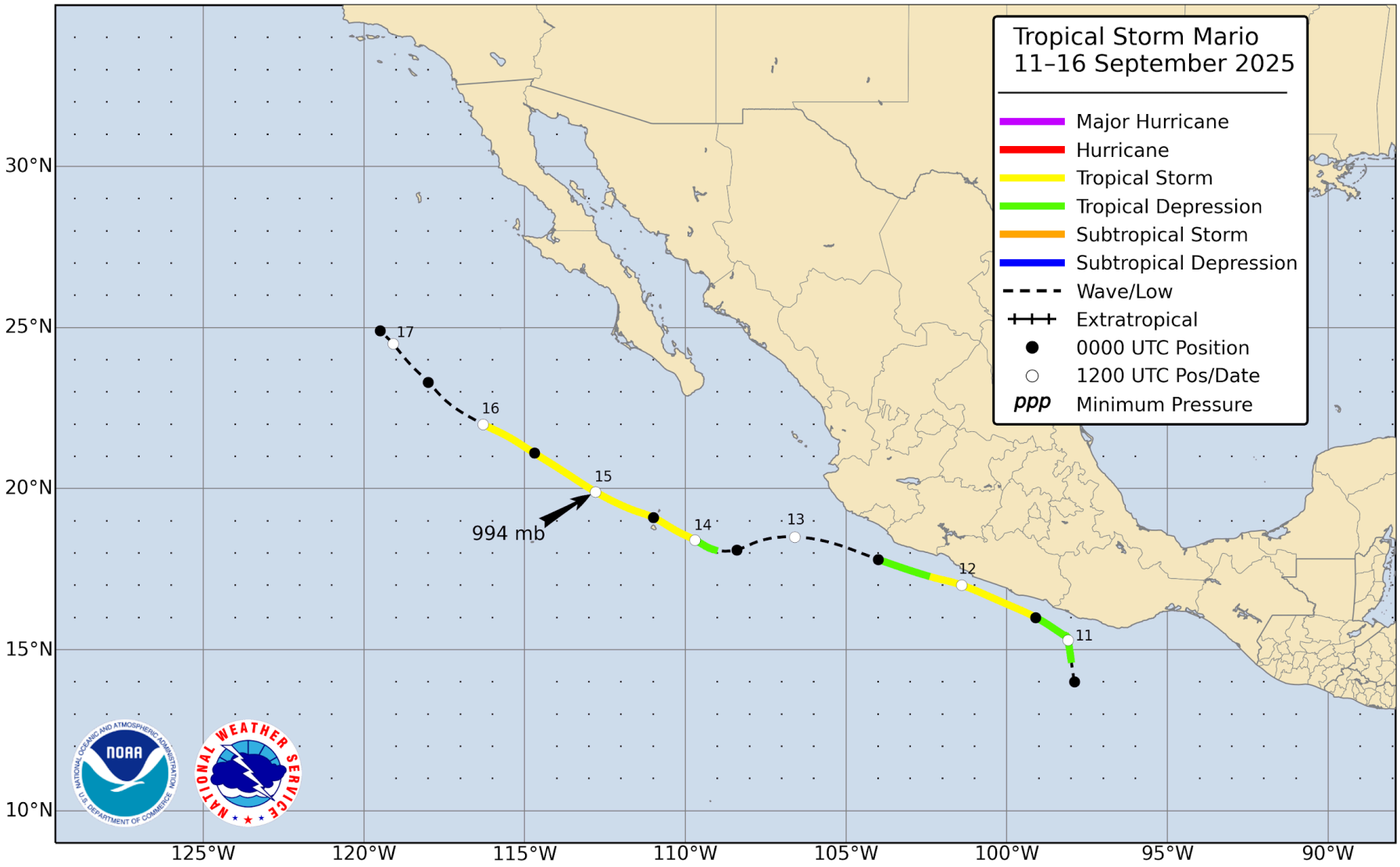


Figure 1. Best track positions for Tropical Storm Mario, 11–16 September 2025.

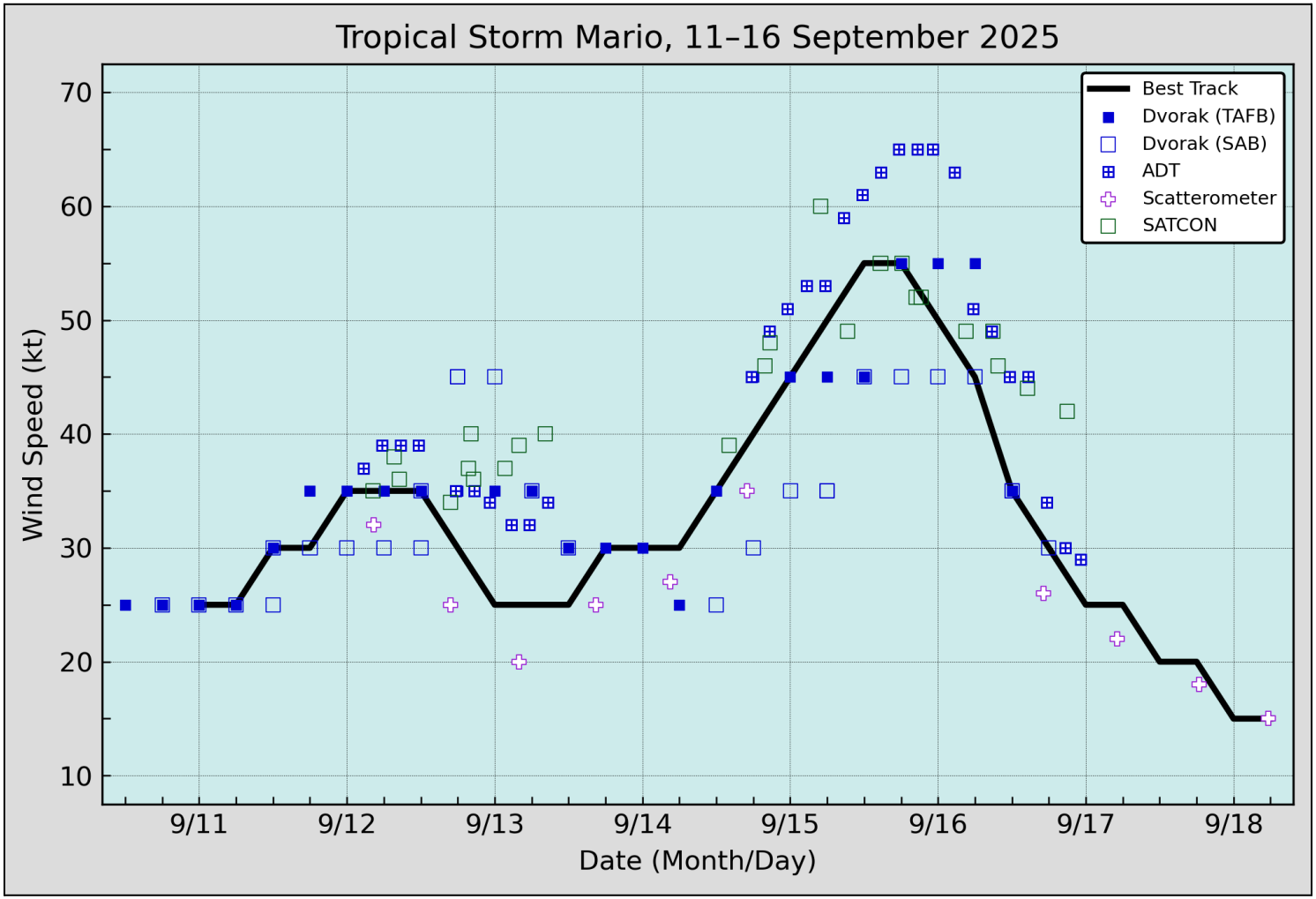


Figure 2. Selected wind observations and best track maximum sustained surface wind speed curve for Tropical Storm Mario, 11–16 September 2025. Advanced Dvorak Technique estimates represent the Current Intensity at the nominal observation time. SATCON intensity estimates are from the Cooperative Institute for Meteorological Satellite Studies. Dashed vertical lines correspond to 0000 UTC.

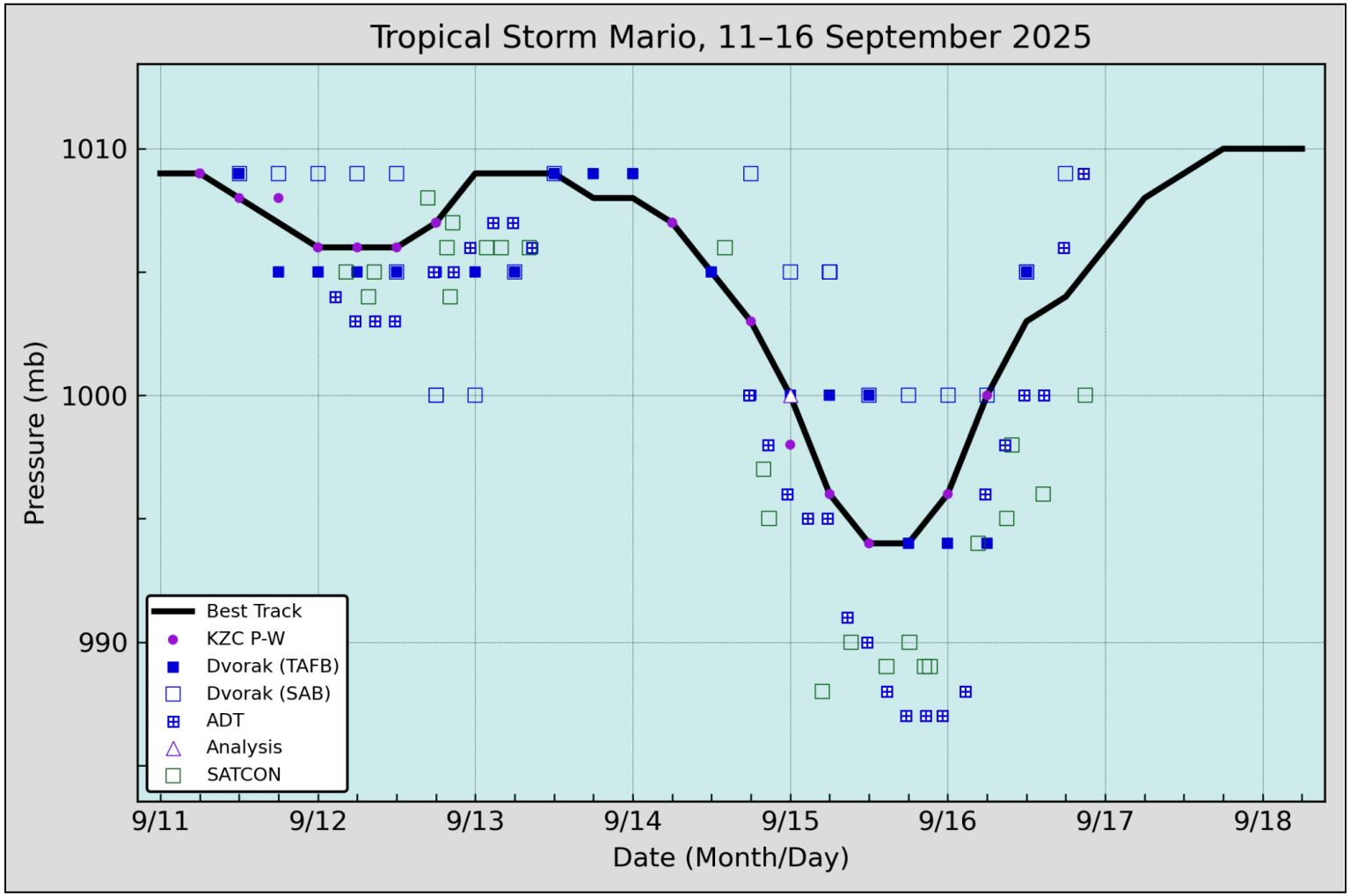


Figure 3. Selected pressure observations and best track minimum central pressure curve for Tropical Storm Mario, 11–16 September 2025. Advanced Dvorak Technique estimates represent the Current Intensity at the nominal observation time. SATCON intensity estimates are from the Cooperative Institute for Meteorological Satellite Studies. KZC P-W refers to pressure estimates derived using the Knaff-Zehr-Courtney pressure-wind relationship. Dashed vertical lines correspond to 0000 UTC.

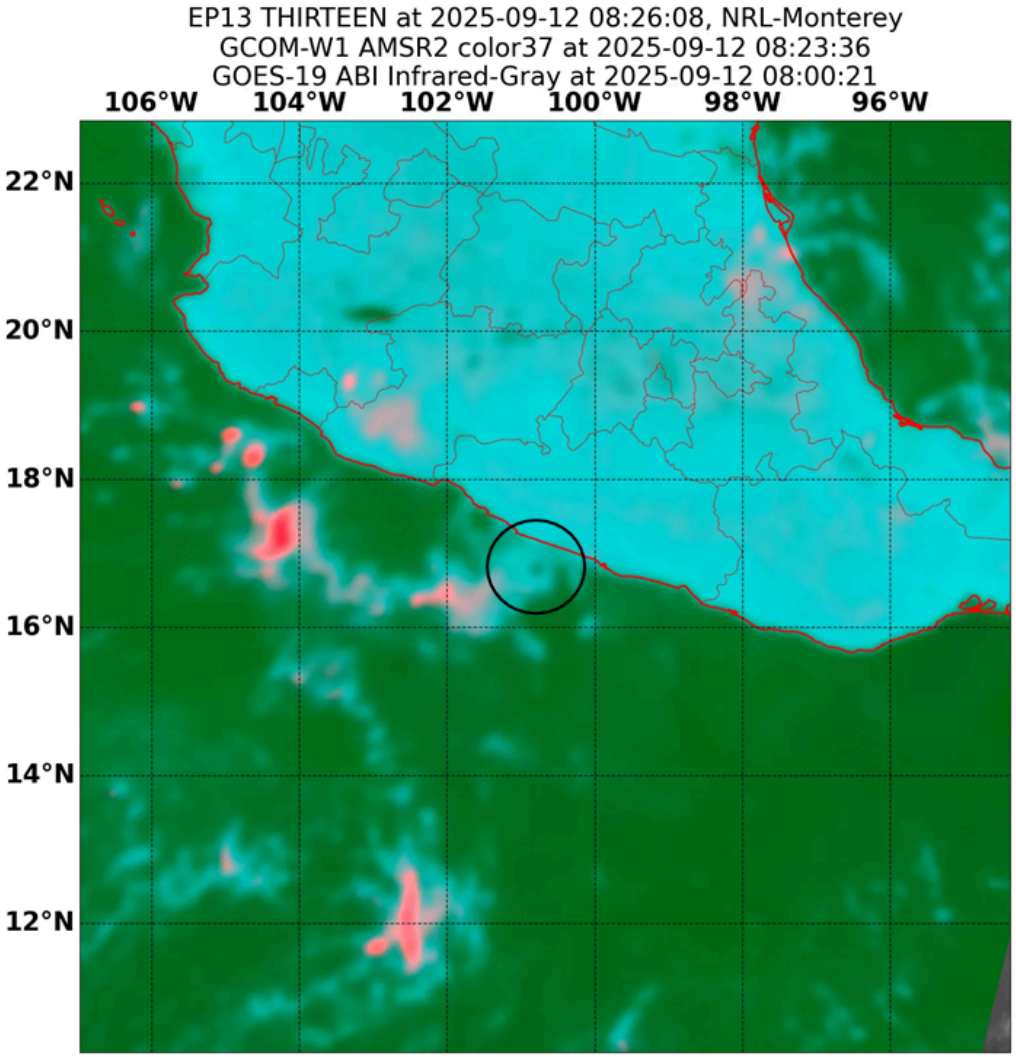


Figure 4. Passive 37 GHz AMSR2 color composite microwave image of Tropical Storm Mario at 0823 UTC 12 September when it was near the coast of southwestern Mexico. The small circulation center of the storm is circled in black. Image courtesy NRL.

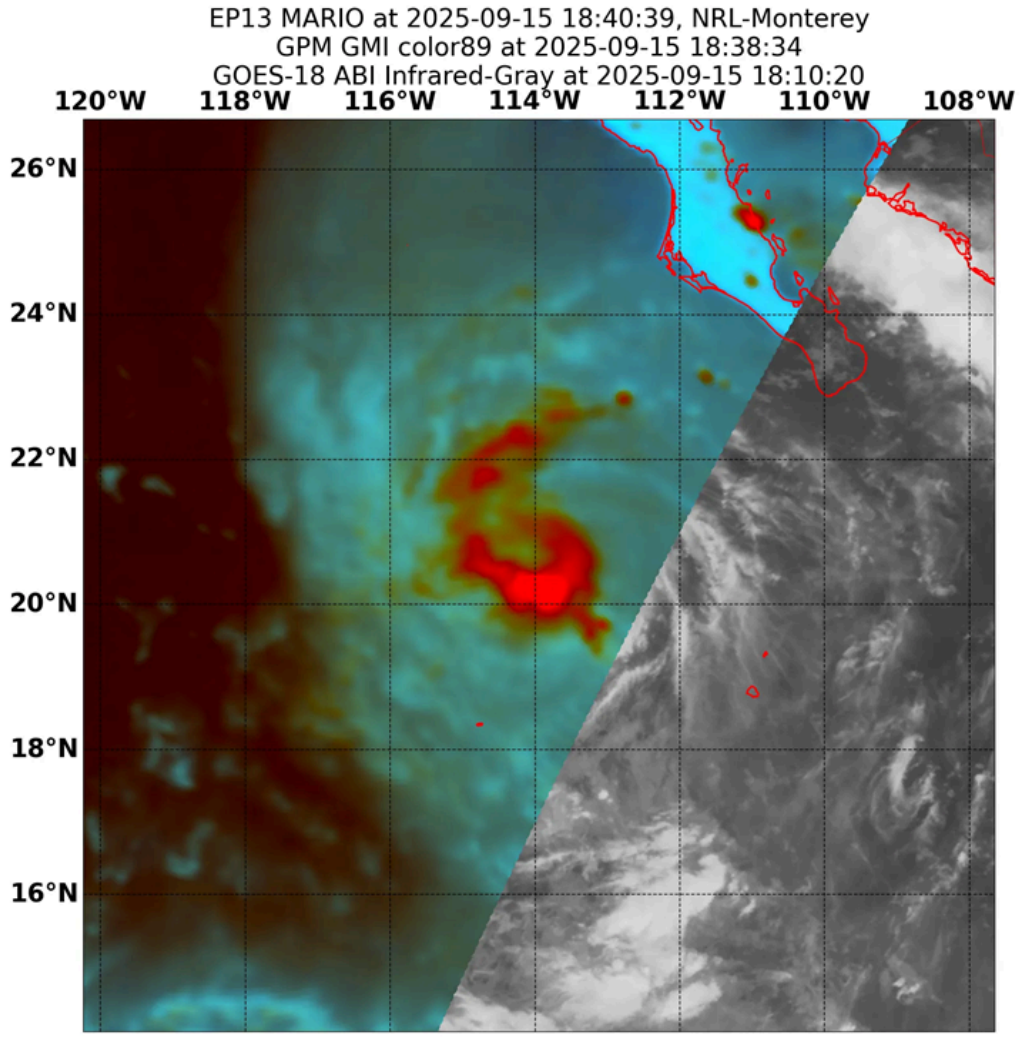


Figure 5. Passive 89 GHz GMI color composite microwave image of Tropical Storm Mario at 1838 UTC 15 September, near the time of its peak intensity. Image courtesy NRL.

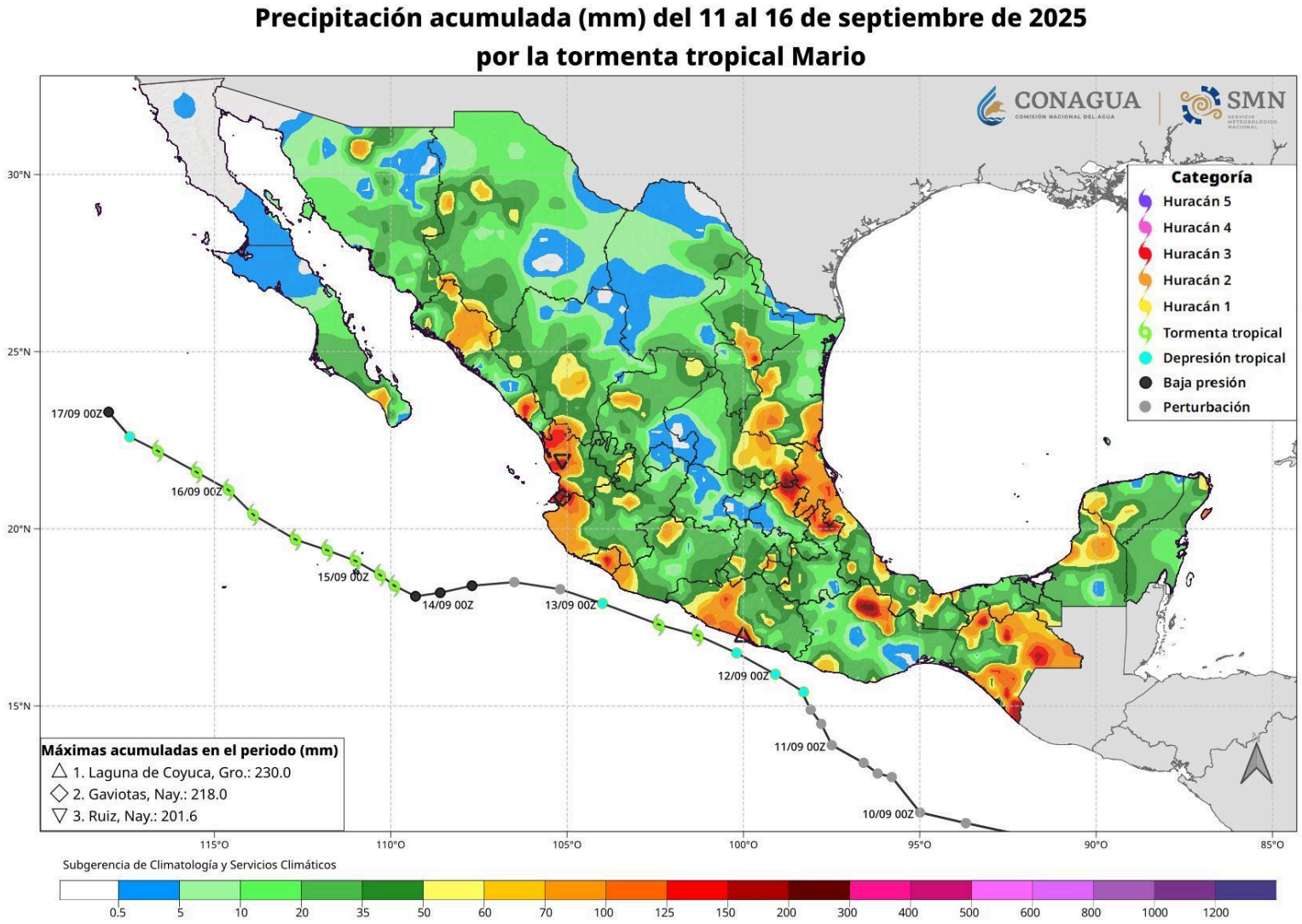


Figure 6. Rainfall accumulations (mm) in Mexico from 11–16 September 2025. Track and intensity are based on the operational NHC assessment. Not all of the rainfall depicted here is directly related to Mario. Image courtesy of CONAGUA, the National Meteorological Service of Mexico.

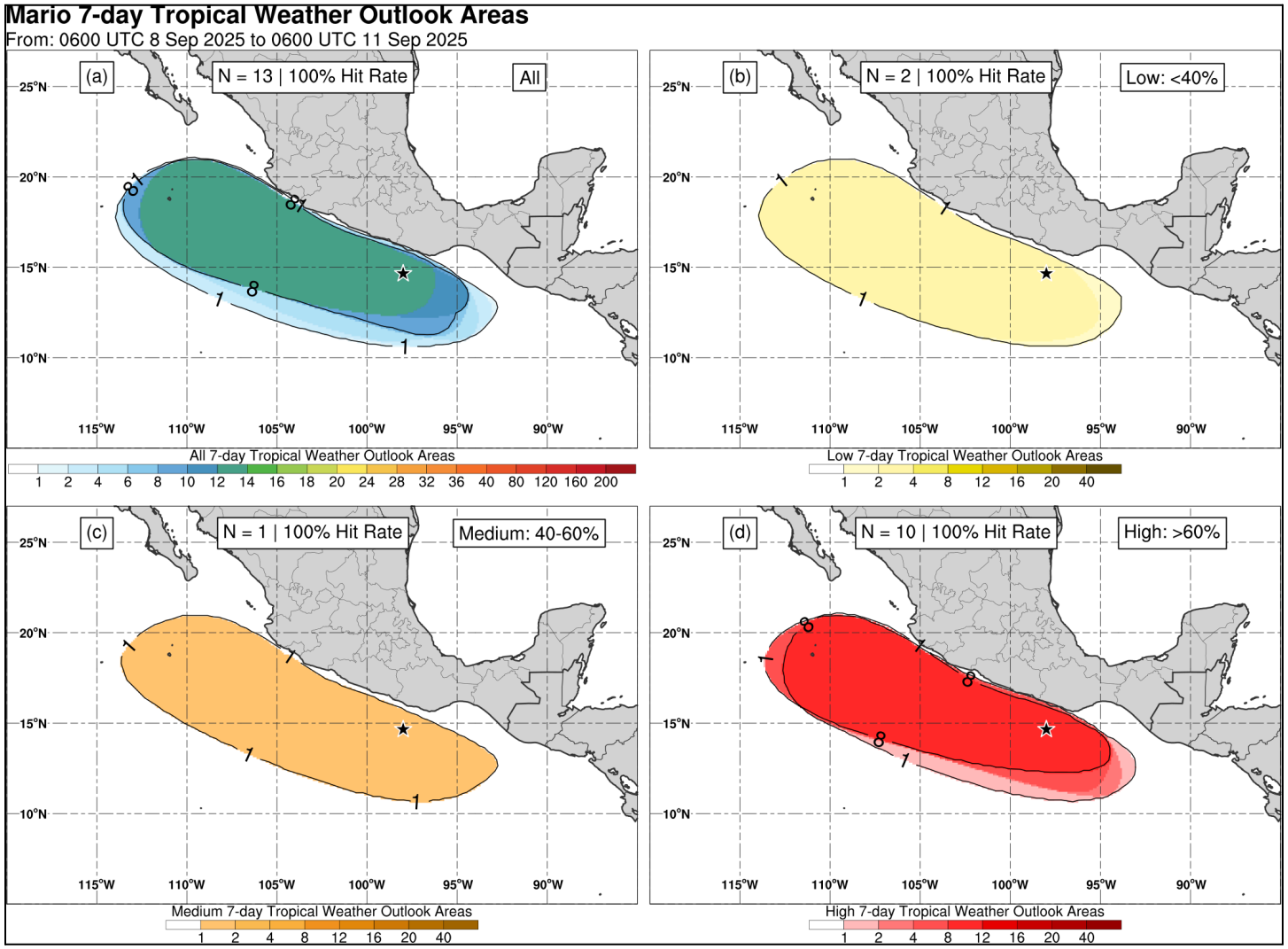


Figure 7. Composites of 7-day tropical cyclone genesis areas depicted in NHC's Tropical Weather Outlooks prior to the formation of Tropical Storm Mario for (a) all probabilistic genesis categories, (b) the low (<40%) category, (c) medium (40–60%) category, and (d) high (>60%) category. The location of genesis is indicated by the black star.

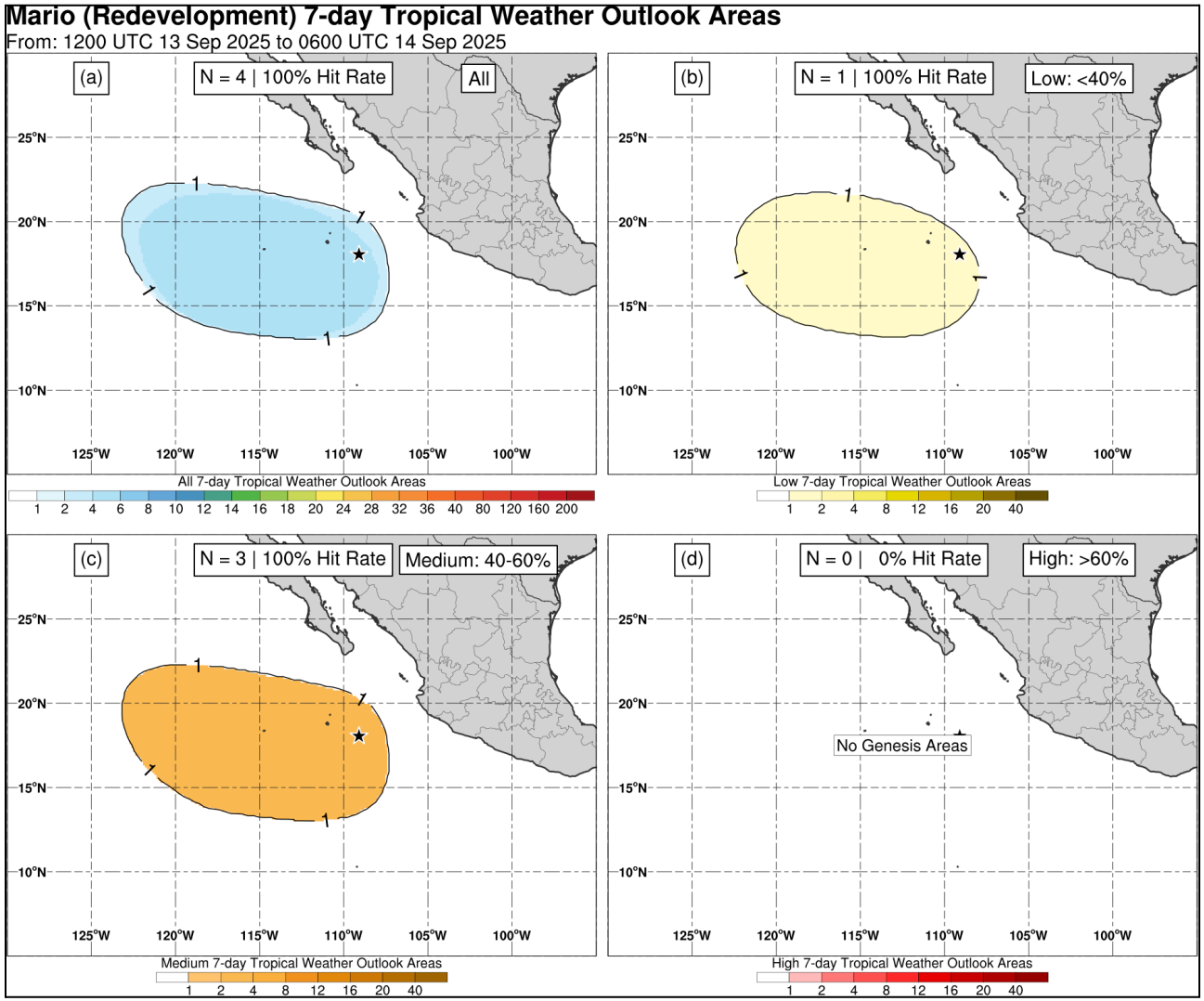


Figure 8. Composites of 7-day tropical cyclone genesis areas depicted in NHC's Tropical Weather Outlooks prior to the redevelopment of Tropical Storm Mario for (a) all probabilistic genesis categories, (b) the low (<40%) category, (c) medium (40–60%) category, and (d) high (>60%) category. The location of genesis is indicated by the black star.